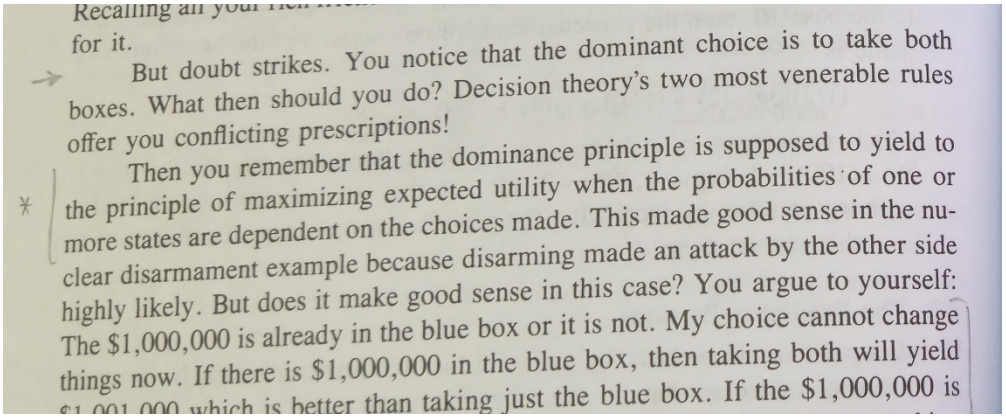
Notes on the predictor paradox

Resnik discusses this paradox in section 4-5, and you should read that whole section (pp. 109 - 111), except you can skip the last paragraph (on smoking). I'll give you a final version of the paradox (the athlete-academic version) at the end of these notes.

When reading Resnik's description of the paradox on pp. 109 – 110, first, keep in mind that the predictor is only a predictor. He or she is ***not*** capable of changing what is in the blue box after you return to the room (and the lid is on the blue box).

Second, it doesn't really matter if the predictor is a *good* predictor, a *very good* predictor, or an *almost perfect* predictor, but most descriptions of this paradox put the probability that predictor has made the correct prediction higher than 90 percent. Often, it's more like 99 percent. This percentage won't change which option has the higher expected value, but it might help to think about the predictor as being very, very good (although not absolutely perfect).

Third, this is in the middle of p. 110:



Recalling all your ...
for it.

But doubt strikes. You notice that the dominant choice is to take both boxes. What then should you do? Decision theory's two most venerable rules offer you conflicting prescriptions!

Then you remember that the dominance principle is supposed to yield to the principle of maximizing expected utility when the probabilities of one or more states are dependent on the choices made. This made good sense in the nuclear disarmament example because disarming made an attack by the other side highly likely. But does it make good sense in this case? You argue to yourself: The $1,000,000 is already in the blue box or it is not. My choice cannot change things now. If there is $1,000,000 in the blue box, then taking both will yield $1,001,000 which is better than taking just the blue box. If the $1,000,000 is

You didn't read the nuclear disarmament example, which is on pp. 9 - 10. It's an example that was often used during the Cold War, and our society doesn't have the same mindset now. But the basic idea is that choosing an option like 'disarm' (or just 'spend significantly less on military preparedness') makes the state 'military attack' more likely than it would be if the option chosen is 'be prepared for a military attack'. Therefore, even when one option dominates another, we might not deploy the dominance principle if the "probabilities of one or more states

are dependent on the choices made" as Resnik says on p. 89. But what does Resnik say in the rest of that paragraph?

The Calvinist version of the predictor paradox

In the middle of p. 111, Resnik explains (briefly) how the Calvinist version of Christianity can give rise to its own predictor paradox. According to Calvinism (or at least this interpretation of it), God has already determined if each of us is going to heaven or to hell after we die. Moreover, it is typically the case that people who are going to heaven are devout (and so don't commit sins) and people who are not going to heaven are not devout (and sometimes sin).

The decision to be made here is about whether or not to commit a sin.

Nothing that you (a "dyed-in-the-wool Calvinist in this account) do now is going to change whether or not you go to heaven, although committing the sin might be evidence that, before you were born, God decided that you were going to hell. (But not committing the sin won't get you into heaven if that wasn't preordained for you. And committing the sin won't send you to hell if that wasn't preordained for you.)

Assuming that going to heaven is preferred to going to hell and committing the sin is preferred to not committing it, here are the outcomes listed from most to least preferred.

outcomes	utility values	probability
commit the sin & go to heaven	1.0	.10
don't commit the sin & go to heaven	__	.90
commit the sin & go to hell	__	.90
don't commit the sin & go to hell	0	.10

Presumably, especially for a "died-in-the-wool Calvinist," the second most preferred outcome is very highly valued. And the third most preferred outcome (commit the sin & go to hell) is not. So, let's set them as .99 and .01. (although as long as the values placed on these two outcomes aren't too close, it doesn't really matter what they are). This gives us the following decision table (with the probability for each outcome given with the utility value for it).

	go to heaven	go to hell
commit the sin	u = 1 (P = .10)	u = .01 (P = .90)
don't commit the sin	u =.99 (P = .90)	u = 0 (P = .10)

It's important to understand that the idea here is ***not*** this: if you commit the sin, then there's a 90 percent chance that God will send you to hell. He's already decided. But if you do commit the sin, that that will indicate that you're a sinner who God has already chosen to go to hell.

What does the dominance principle instruct you to do in this case?

What does the expected utility strategy instruct you to do?

The athlete-academic version of the predictor paradox

This is not in section 4-5. Read it instead of the paragraph that begins, "Now for a contemporary example."

Jeff knows that either Sam or Tom is his father, but he does not know which. Sam died of a very painful inherited disease that strikes in one's middle thirties and Tom did not. The disease is genetically dominant, and Sam carried only the dominant gene. Tom did not have the gene. If Sam is his father, the Jeff will die of the dreaded disease. If Tom is his father, he will not. Furthermore, suppose there is a well-confirmed theory that states a person who inherits this gene will also inherit a tendency toward behavior that is characteristic of intellectuals and scholars. Sam had this tendency. Neither Tom nor the person's mother had such a tendency.

Jeff is now deciding whether to go to graduate school or to become a professional baseball player. He prefers (although not enormously) the life of an academic to that of a professional athlete. Regardless of whether or not he will die in his middle thirties, he would be happier as an academic. The choice of the academic life would thus appear to be his best choice.

Now suppose he reasons that if he decides to be an academic, the decision will show that he has such a tendency and therefore it will be likely that he carries the gene for the disease and will die in his middle thirties. On the other hand, if he chooses to become a baseball player, it will be likely that Tom is his father; therefore, he is not likely to die of the disease.[1]

What should he do? Here is the decision table.

	doesn't have the gene	**has the gene**
go to grad school (& become an academic)	u = 1 (P = .10)	u = .01 (P = .90)
enter the MLB draft (& become a professional athlete)	u =.99 (P = .90)	u = 0 (P = .10)

What does the dominance principle instruct Jeff to do in this case?

What does the expected value strategy instruct him to do?

[1] from Gardner, M. (1974). Mathematical Games. In *Scientific American*, pp. 103.